Reading a Simple Linear Regression

Jinsong Chen

Table of contents

Chapter box

Central question: How can we read a fitted regression line, explain its calculation and distinguish its descriptive answer from model-based inference?
Focal concepts: Slope, intercept, fitted value and residual; least squares and covariance; residual SD and R-squared; slope SE and confidence interval; repeated sampling and precision; mean-response and new-case intervals; form, spread, influence and support; reading a reported argument before evaluating it.
Main evidence: The 102 Canadian occupations in `carData::Prestige`, with income in thousand 1971 Canadian dollars, plus small declared teaching examples and a simulation with known coefficients.
Scope: Chapters 2–4 supply statistical foundations and Chapters 5–6 supply elementary R. Chapter 1 supplies the argument vocabulary; Chapter 1a is optional. The nine sections retain the ordinary regression method and its analytical explanations. Appendices A–B add extended derivations, matrix covariance and a projection target. The lab supplies operating practice; Chapter 8 adds adjustment and the full Project STAR design transfer.

Target map. Sections 1–3 stay with the descriptive least-squares slope for the 102 fixed occupation records. Sections 4–5 explicitly open a separate hypothetical process question whose estimand is a constant conditional-mean slope, β_1. Section 6 distinguishes a fitted mean at a specified education value from a new occupation's outcome; Section 7 checks form, spread, influence, and support. Section 8 then reads and evaluates the chapter's reported finite-record argument, and Section 9 summarizes the route. This map prevents a model-based interval from being silently attached to the descriptive answer.

1 A Question About Occupations and a Line Through Their Records

1.1 Begin with the unit and the question

Suppose a report says, "Another year of education increases income by about 900 dollars." Before examining its significance, ask whose education and whose income were measured. The same number could refer to individuals, households, or occupational averages. Those are different comparisons.

Our question is deliberately narrower:

> Among the 102 recorded Canadian occupations, how did occupation-average income differ across the supported range of occupation-average education in 1971, as summarized by a fitted straight line?

One row represents an **occupation**, not a worker. An occupation with higher average education need not contain the same people as an occupation with lower average education. A comparison between their averages does not describe what would happen to one person's income after another year of schooling. Aggregate and individual relationships can differ substantially (Robinson, 1950).

The data are distributed as *carData::Prestige* (Fox et al., 2022; Fox & Weisberg, 2019). The package documentation attributes education and income to the *Census of Canada* (1971), volume 3, part 6, pages 19-1 to 19-21. We read the package documentation as a source record, not as evidence that these occupations form a probability sample. The chapter's calculations and teaching report are our reanalysis, not statements attributed to an original research paper.

Table 1
Record, measurements, and scope for the focal analysis.

Feature	Meaning in this chapter
Unit and record count	102 occupations; each occupation receives equal weight
Education, x	Average years of education of occupational incumbents in 1971
Income, y	Average income of incumbents, divided by 1,000; thousand 1971 Canadian dollars
Missingness for the fit	No missing education or income values; no modeled rows excluded
Observed support	Education: 6.38–15.97 years; income: 0.611–25.879 thousand dollars
Descriptive estimand	Least-squares slope summarizing these 102 fixed occupation records
Limits of the source record	No probability-sampling basis or individual-level analysis established here

Equal weighting matters. A small occupation and a large occupation contribute one row each. Without suitable counts and a different estimand, this analysis does not describe an average worker's experience. Historical dollars also cannot be read as contemporary purchasing power.

1.2 Look before fitting

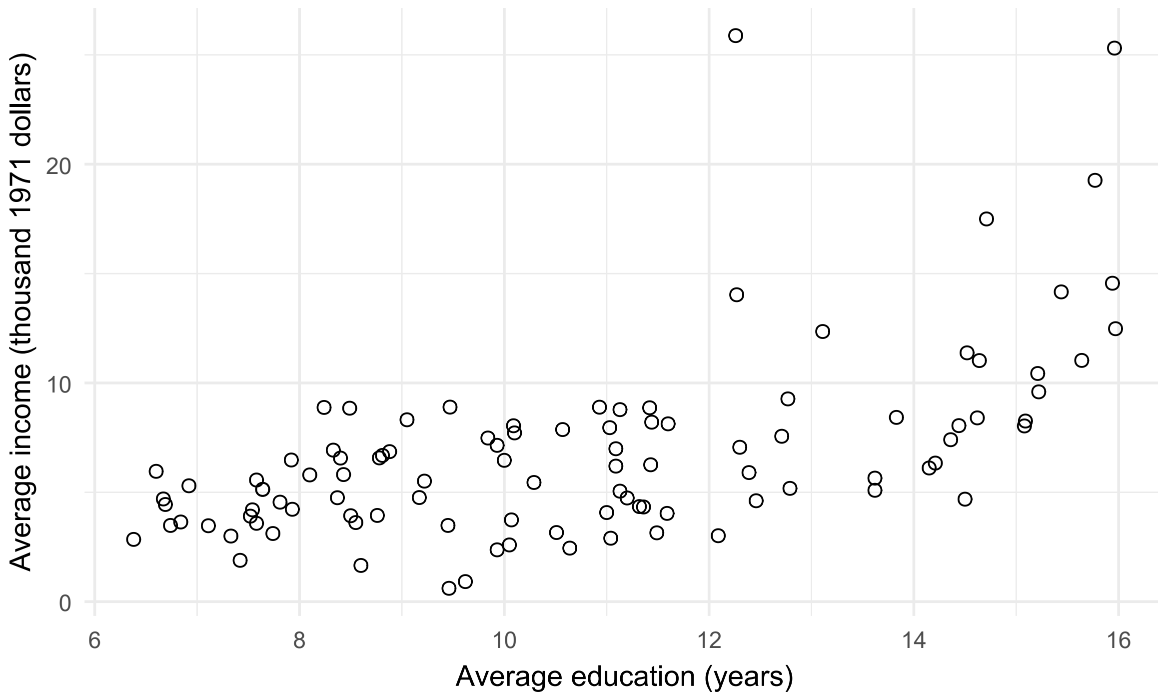

Figure 1: Education and income in the 102 occupation records, before fitting a line.

The scatterplot gives the line a job: summarize an overall upward pattern amid substantial variation. It also shows why the line cannot describe every occupation equally well. General managers, for example, have an average education of 12.26 years and an average income of 25.879 thousand dollars; university teachers have corresponding values of 15.97 and 12.480. More education does not order every pair of occupations by income.

Here is the short R preparation used throughout. Package installation and project setup are developed in the computing chapters. The data come from the named `carData` package, so this analysis requires no separate classroom file.

```r
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
library(ggplot2)
ggplot(prestige, aes(education, income_k)) + geom_point() +
  labs(x = "Mean education (years)", y = "Mean income (thousand dollars)")
model <- lm(income_k ~ education, data = prestige)
x <- prestige$education
y <- prestige$income_k
b <- coef(model)
coef(model)
```

The formula puts the outcome to the left of the tilde and the predictor to its right. The command finds coefficients. It does not choose a population, establish a causal design, or justify a confidence interval.

2 Read the Fitted Line and Understand How OLS Finds It

2.1 Fitted values, slopes, and residuals

A straight-line summary assigns each observed predictor value x_i a **fitted value** $\hat{y}_i$:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad e_i = y_i - \hat{y}_i.$$

The subscript i indexes a record. The hat marks a fitted quantity. The **residual** e_i is observed income minus fitted income. A positive residual places an occupation above the line; a negative residual places it below. A residual is calculated after fitting and is not the same as an unobservable error from a population model.

For the occupation records, the line is

$$\widehat{\text{income}} = -2.8536 + 0.8988\,\text{education}.$$

Income is still in thousands of dollars. The slope says that the fitted income differs by 0.8988 thousand dollars, about 899 dollars, for a one-year difference in occupation-average education. A two-year difference along this line corresponds to $2(0.8988) = 1.7976$ thousand dollars. These are comparisons on the fitted line, not predicted gains from changing a worker's education.

For university teachers,

$$\hat{y} = -2.8536 + 0.8988(15.97) = 11.5002, \qquad e = 12.4800 - 11.5002 = 0.9798.$$

Their recorded average income is about 980 dollars above the fitted value. Using the unrounded coefficients gives essentially the same rounded answer. For an occupation averaging 12 years of education, the fitted value is $-2.8536 + 0.8988(12) = 7.9320$ thousand dollars.

The intercept places the line vertically: it is its fitted income at education zero. Zero is outside the observed range, so the negative intercept is not evidence that any recorded occupation has negative average income. Centering education at 12 would change the intercept to 7.9322 while leaving the slope and all fitted values unchanged.

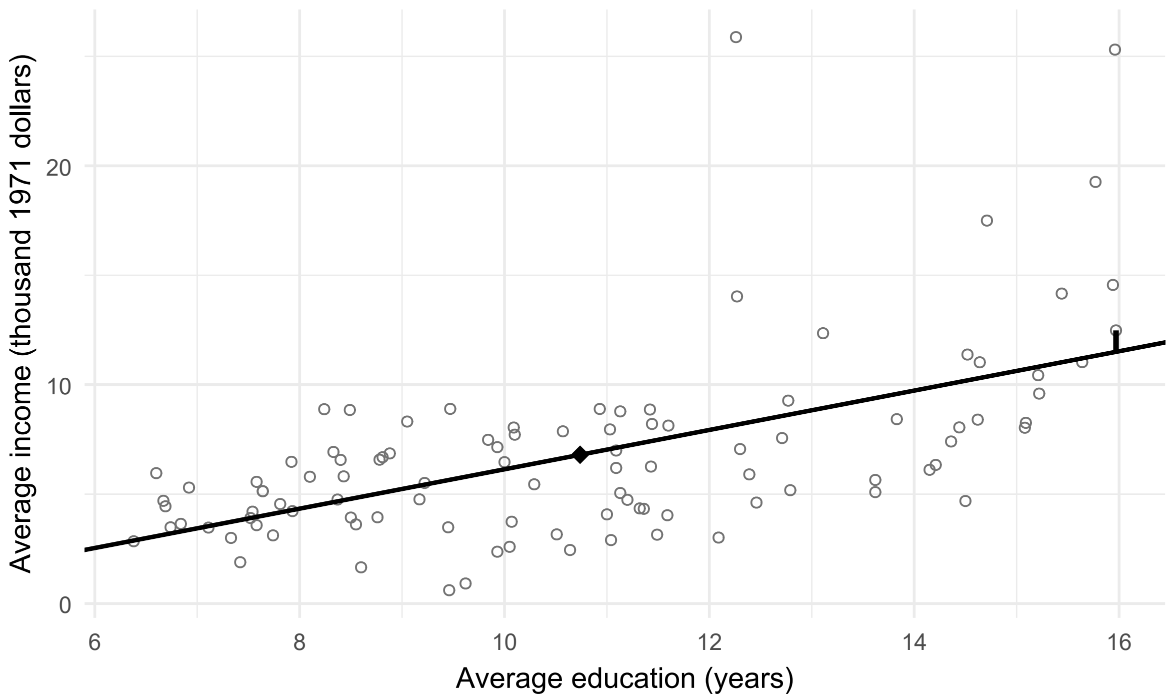

Figure 2: The least-squares line passes through the mean point; the marked vertical segment is the university-teacher residual.

2.2 Why this particular line?

Ordinary least squares (**OLS**) chooses the intercept and slope that minimize the **residual sum of squares**:

$$RSS = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

The summation sign says to add the squared residual from every record. Squaring prevents positive and negative deviations from cancelling and gives large deviations more influence on the criterion. "Best" here means best by this criterion among straight lines with an intercept, not best for every research purpose.

Two centered sums lead directly to the solution:

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \qquad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}),$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Here $\bar{x}, \bar{y}$ are means; s_x, s_y are sample standard deviations; s_{xy} is sample covariance; and r is Pearson's correlation. The variance and covariance both divide their centered sums by $n-1$, so that divisor cancels in their ratio. The slope has income-per-education units; correlation has no units.

For these data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} = 0.8988, \qquad \hat{\beta}_0 = 6.7979 - (0.8988)(10.7380) \approx -2.8534.$$

The final small rounding difference is avoided by using unrounded summaries. Positive covariance produces a positive slope, provided education varies at all. If every occupation had the same education, $S_{xx} = 0$ and a slope could not be identified.

The fitted line passes through $(\bar{x}, \bar{y})$, because substituting $\bar{x}$ into the intercept formula gives $\bar{y}$. With an intercept, the residuals also sum to zero and their products with x sum to zero. Otherwise the line could be shifted or tilted to reduce RSS. These are algebraic properties of OLS, not tests of independence or linearity. Full normal-equation and projection-matrix derivations are retained in **Appendix A**.

2.3 A tiny calculation you can inspect completely

Consider the teaching records $(x, y) = (1, 2), (2, 4), (3, 3)$. Their means are $\bar{x} = 2, \bar{y} = 3$, so

$$S_{xx} = (-1)^2 + 0^2 + 1^2 = 2, \qquad S_{xy} = (-1)(-1) + (0)(1) + (1)(0) = 1.$$

Thus the slope is $1/2 = 0.5$, the intercept is $3 - 0.5(2) = 2$, and the fitted values are 2.5, 3, and 3.5. The residuals are $-0.5, 1, -0.5$, which sum to zero, and $RSS = 0.25 + 1 + 0.25 = 1.5$. No normality assumption is needed for these calculations. A stochastic model becomes necessary when we ask about repeated-sample uncertainty.

2.4 Compare candidate lines with the OLS minimum

The least-squares criterion also lets us compare the fitted line with alternatives. To keep the comparison transparent, make every candidate line pass through $(\bar{x}, \bar{y})$ and vary only its slope. For such a line,

$$RSS(b_1) = RSS(\hat{\beta}_1) + S_{xx}(b_1 - \hat{\beta}_1)^2.$$

The added term is zero only at the OLS slope and positive for every other slope when $S_{xx} > 0$. The following table checks five deliberately chosen lines on the occupation records. These are alternative summaries of the same records, not repeated samples.

```
candidate_slopes <- unname(b[2]) + c(-0.6, -0.3, 0, 0.3, 0.6)
candidate_intercepts <- mean(y) - candidate_slopes * mean(x)
ols_rss <- sum(resid(model)^2)
predictor_spread <- sum((x - mean(x))^2)
candidate_rss <- ols_rss +
  predictor_spread * (candidate_slopes - b[2])^2
candidate_lines <- data.frame(
  Intercept = candidate_intercepts,
  Slope = candidate_slopes,
  RSS = candidate_rss
)
```

Each operation follows the displayed identity. R applies the same arithmetic to all five candidate slopes, so no function or loop is needed here.

Table 2
Candidate lines through the mean point and their residual sums of squares.

Intercept	Slope	RSS
3.589	0.299	1484
0.368	0.599	1281
-2.854	0.899	1213
-6.075	1.199	1281
-9.296	1.499	1484

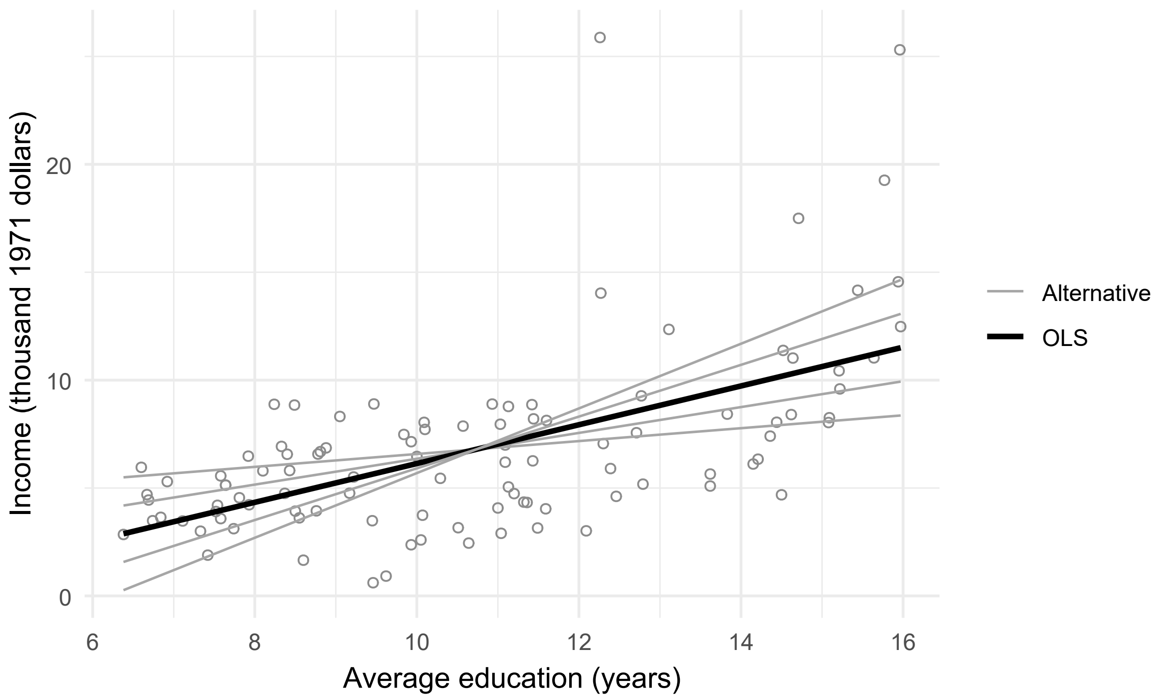

Figure 3: Five candidate lines through the sample mean. The OLS line has the smallest residual sum of squares.

The comparison also explains why high-income occupations can exert substantial influence: squaring gives large vertical discrepancies extra weight. That is a feature of the criterion, not evidence that those occupations are invalid.

2.5 Centering changes the reference point, not the line

The fitted equation can be written

$$\hat{y}_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x}).$$

Centering education at its observed mean makes the intercept equal to $\bar{y} = 6.7979$ thousand dollars, the fitted income at average education. It leaves the slope, fitted values, residuals, RSS, and R^2 unchanged. Centering at 12 instead makes the intercept the already computed fitted value 7.9322. Thus centering can make the intercept interpretable without changing the comparison represented by the line.

```r
prestige$education_c <- prestige$education - mean(prestige$education)
model_centered <- lm(income_k ~ education_c, data = prestige)
coef(model_centered)
max(abs(fitted(model_centered) - fitted(model)))
```

Changing income from thousands of dollars to dollars would multiply the intercept, slope, residual SD, and coefficient SEs by 1,000, while leaving t statistics, p values, correlation, and R^2 unchanged. A unit change is not new evidence or a stronger relationship.

3 Describe Fit Before Making Model-Based Inferences

3.1 How much variation remains?

A fitted line can show a clear linear trend and still leave large differences between occupations unexplained by its fitted values.

For OLS with an intercept,

$$TSS = \sum_i (y_i - \bar{y})^2 = RegSS + RSS, \qquad RegSS = \sum_i (\hat{y}_i - \bar{y})^2.$$

For the occupation records, $1820.8134 = 607.4214 + 1213.3920$. This is a decomposition of squared deviations, in squared thousand-dollar units. It holds because the cross-product of fitted deviations and residuals is zero.

The coefficient of determination is

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{1213.3920}{1820.8134} = 0.3336.$$

The fitted line accounts for about 33.4% of the variation in recorded income around its mean, according to this squared-deviation measure. "Accounts for" or "explains" in this algebraic sense does not mean causes. For simple regression with an intercept, $R^2 = r^2$: $0.5776^2 \approx 0.3336$. The identity need not hold for other models or a no-intercept fit.

The residual standard deviation is

$$s_e = \sqrt{\frac{RSS}{n-2}} = \sqrt{\frac{1213.3920}{100}} = 3.4834.$$

Its units are thousand dollars, the same as income. Fitting an intercept and slope imposes two independent residual constraints, leaving $n - 2$ residual degrees of freedom. Under the constant-variance model introduced next, s_e estimates the error standard deviation. Without that model it remains a scaled measure of residual scatter; it is not the standard error of the slope or a guaranteed typical prediction error in new records.

Table 3
Different summaries have different units and jobs.

Quantity	Occupation result	What it describes
Slope	0.8988	Change in fitted thousand-dollar income per education year
Residual SD	3.4834	Remaining income scatter on the outcome scale
R^2	0.3336	Proportion of centered squared variation accounted for in these records
Classical slope SE	0.1270	Model-based repetition variability of the slope, under the next assumptions

4 Make and Read a Model-Based Inference

4.1 Announce the inferential extension before its interval

The finite-record slope is not waiting to be discovered in a larger population: we have computed it for all 102 records named in the focal question. A confidence interval answers an additional question.

For a **separable teaching extension**, suppose occupations' outcomes arose independently from a process with conditional mean

$$E(Y_i \mid X_i = x_i) = \beta_0 + \beta_1 x_i.$$

The estimand β_1 is now a constant conditional-mean slope in that hypothetical process, not the fixed slope of these 102 records. Repetition means generating new outcomes from that process at the stated predictor values and refitting. It does not mean that the documented occupations were randomly sampled. Changing the estimand changes the question–answer pair.

Write the process as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad E(\varepsilon_i \mid X) = 0, \qquad \mathrm{Var}(\varepsilon_i \mid X) = \sigma^2.$$

For the exact classical t procedure, errors are independent and conditionally normal with this common variance, and $S_{xx} > 0$. The x_i values need not be normal; neither must the marginal distribution of Y. It is the conditional error distribution, not the histogram of every variable, that appears in the model. Independence is supported by how records arise, not by a good-looking residual plot.

To see where slope uncertainty comes from, substitute the process equation into the OLS slope formula. Here X denotes the predictor values on which we condition. Since $\sum_i (x_i - \bar{x}) = 0$,

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x})\varepsilon_i}{S_{xx}}.$$

The fitted slope differs from the process slope because the realized errors need not cancel when weighted by distance from mean education. Independence makes their covariance terms zero; common variance gives

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2 \sum_i (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

This variance identity does not require normal errors. Normality enters the exact t reference. Taking the square root and estimating σ by s_e gives the slope standard error:

$$SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{S_{xx}}} = \frac{3.4834}{\sqrt{751.8852}} = 0.1270,$$

$$\hat{\beta}_1 \ \pm \ t_{.975,n-2} SE(\hat{\beta}_1) = 0.8988 \pm 1.984(0.1270) = [0.6468, \ 1.1508].$$

The critical value 1.984 is the 97.5th percentile of a t distribution with 100 degrees of freedom. More error variation raises the SE; greater predictor spread lowers it when the other quantities are held fixed. More observations usually improve precision through their contribution to S_{xx}, but dependent observations do not supply independent information merely by increasing the row count.

The null test $H_0 : \beta_1 = 0$ uses $t = 0.8988/0.1270 = 7.075$ and a two-sided $p < .001$. This is evidence against that zero-slope model under the stated procedure. It is not the probability that the null is true, the probability that the data occurred by chance, evidence of causation, or a measure of practical importance. The interval describes the range produced by this uncertainty method, not a 95% probability assigned to the fixed parameter after observing this interval. The next section makes the repeated-sample interpretation visible before prediction is introduced.

The assumptions are a proposed justification, not facts created by the formula. Curvature, changing spread, or dependence can invalidate the exact coverage claim. The diagnostic section teaches the checks before the reported argument is evaluated.

5 See Sampling Uncertainty Through Repetition

5.1 A known-truth line and one fitted sample

Generate 20 outcomes at predictor values equally spaced from 6 to 16:

$$Y_i = 2 + 0.5x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} \mathcal{N}(0, 1).$$

These units and values are a teaching construction, not occupation records. The true intercept is 2, true slope is 0.5, and true error SD is 1.

```r
set.seed(123)
known_data <- data.frame(x = seq(6, 16, length.out = 20))
known_data$y <- 2 + 0.5 * known_data$x + rnorm(20, sd = 1)
known_model <- lm(y ~ x, data = known_data)
coef(known_model)
sigma(known_model)
```

`coef()` returns the fitted intercept and slope; `sigma()` returns the residual SD. The table adds the coefficient SEs from `summary(known_model)`.

Table 4
One fitted sample from a process with known coefficients and error SD.

Quantity	Estimate	Model SE
Intercept	2.477	0.836
Slope	0.470	0.073
Residual SD	0.995	

The last cell is blank because residual SD is a scale estimate, not a coefficient with a coefficient standard error.

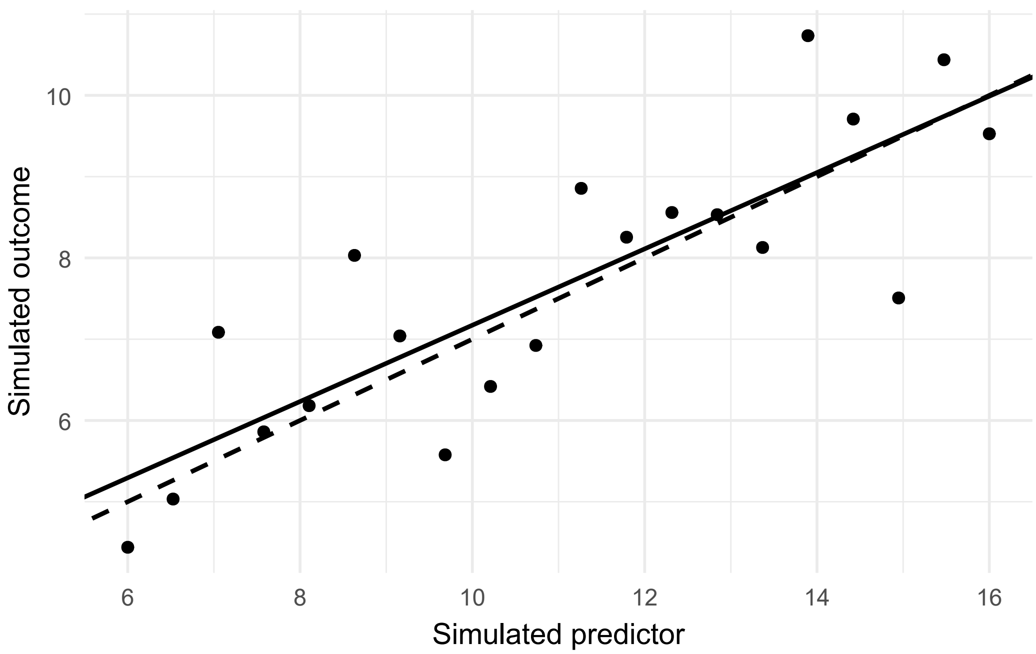

Figure 4: Known and fitted lines for one simulated dataset. The true process is programmed; the fitted line responds to the realized errors.

The fitted slope and intercept need not equal their programmed values in one sample. The unobserved error is the deviation from the true mean; the residual is the observed deviation from the fitted line. Even when the errors are independent, fitted residuals are linked by their zero-sum and zero-covariance constraints. A residual plot is an imperfect diagnostic window on error behavior, not a direct sample of observed true errors.

5.2 Repeated samples and the sources of slope precision

The next experiment holds 40 predictor values fixed and regenerates only the errors, 2,000 times. It records slopes, estimated standard errors, and whether each classical interval covers the known slope 0.5.

```r
set.seed(321)
rep_x <- seq(6, 16, length.out = 40)
repetitions <- data.frame(
  slope = numeric(2000), se = numeric(2000), covered = logical(2000)
)
for (j in seq_len(nrow(repetitions))) {
  rep_y <- 2 + 0.5 * rep_x + rnorm(40, sd = 1)
  rep_fit <- lm(rep_y ~ rep_x)
  repetitions$slope[j] <- coef(rep_fit)[2]
  repetitions$se[j] <- coef(summary(rep_fit))[2, "Std. Error"]
  rep_ci <- confint(rep_fit)[2, ]
  repetitions$covered[j] <- rep_ci[1] <= 0.5 &&
    rep_ci[2] >= 0.5
}
repetition_summary <- c(
  mean_slope = mean(repetitions$slope),
  empirical_SD = sd(repetitions$slope),
  mean_estimated_SE = mean(repetitions$se),
  theoretical_SD = 1 / sqrt(sum((rep_x - mean(rep_x))^2)),
  empirical_coverage = mean(repetitions$covered)
)
```

Table 5

Slope behavior across 2,000 repetitions of the known process.

Quantity	Value
Mean estimated slope	0.5018
SD of estimated slopes	0.0528
Mean estimated SE	0.0531
Theoretical slope SD	0.0534
Fraction of intervals covering 0.5	0.9455

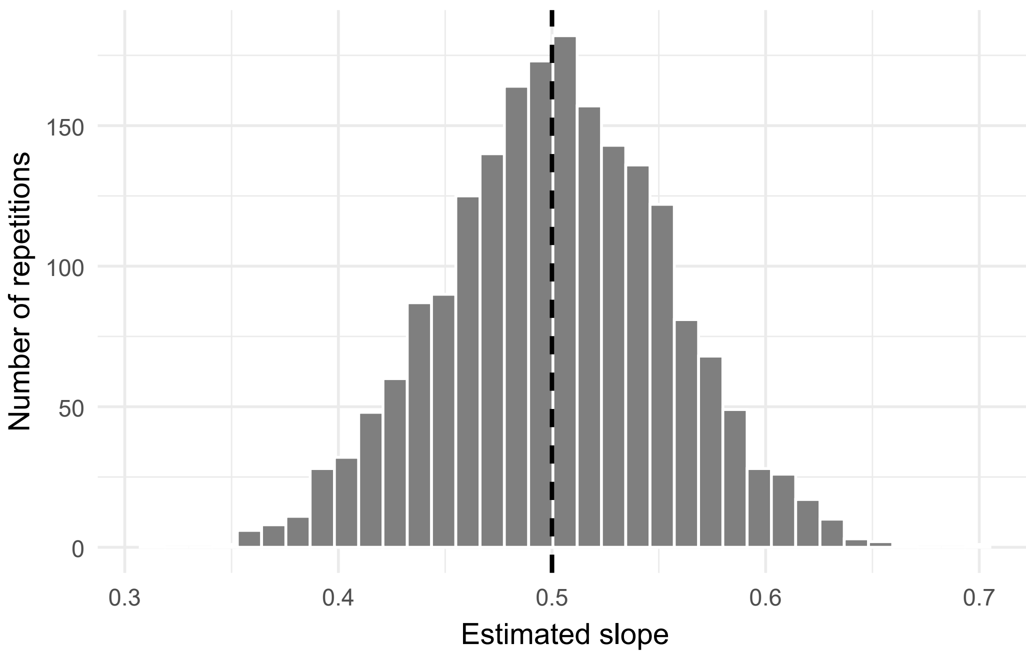

Figure 5: Slopes in 2,000 repetitions of the known process. The vertical reference is the true slope, 0.5.

The empirical SD of the slopes estimates their repeated-sample spread. Each fitted SE tries to estimate that spread using just one sample. Coverage is a property of the repeated interval-producing procedure, not a probability assigned to a fixed parameter after one interval is realized. The empirical coverage will not be exactly .95: at true .95 coverage, its Monte Carlo SE with 2,000 independent repetitions is $\sqrt{.95(.05)/2000} \approx .0049$.

The theoretical slope SD is $\sigma/\sqrt{S_{xx}}$. More noise increases it; greater predictor spread decreases it. More observations typically decrease it when the predictor distribution is held comparable. To isolate each feature, change one at a time:

```r
baseline_x <- seq(6, 16, length.out = 40)
more_x <- seq(6, 16, length.out = 160)
narrow_x <- seq(8.5, 13.5, length.out = 40)
baseline_sd <- 1 / sqrt(sum((baseline_x - mean(baseline_x))^2))
more_rows_sd <- 1 / sqrt(sum((more_x - mean(more_x))^2))
narrow_sd <- 1 / sqrt(sum((narrow_x - mean(narrow_x))^2))
more_noise_sd <- 2 / sqrt(sum((baseline_x - mean(baseline_x))^2))
c(baseline = baseline_sd, more_rows = more_rows_sd,
  narrower_support = narrow_sd, more_noise = more_noise_sd)
```

The first three lines define the predictor grids. Each subsequent line substitutes a noise SD and a grid's spread into the same formula.

Table 6

How sample size, predictor support, and error SD affect theoretical slope precision.

Condition	n	Lower	Upper	Error SD	Slope SD
Baseline	40	6.0	16.0	1	0.0534
More rows	160	6.0	16.0	1	0.0272
Narrower support	40	8.5	13.5	1	0.1068
More noise	40	6.0	16.0	2	0.1068

Halving the predictor range at the same sample size doubles the slope SD; doubling σ also doubles it. Quadrupling the number of equally spaced points over the same range approximately halves it; the small difference from an exact half comes from the finite grids' variances. Using `x <- 1:n` while increasing n would change both sample size and predictor range. More rows do not automatically remove confounding, measurement error, dependence, or misspecification. These simulations establish behavior only under the mechanisms programmed into them.

6 Distinguish a Mean Response from a New-Case Prediction

Slope uncertainty concerns one process coefficient. Prediction introduces a different distinction: estimating a conditional mean versus anticipating one new outcome from the same declared process.

6.1 A fitted mean is not a new occupation

At $x_0 = 12$, the fitted income is 7.9322 thousand dollars. Under the model, this estimates the conditional mean for occupations with that average education. The standard error for the fitted mean is

$$SE(\hat{y}_0) = s_e\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

Substitution gives

$$3.4834\sqrt{\frac{1}{102} + \frac{(12 - 10.7380)^2}{751.8852}} = 0.3803.$$

The resulting 95% mean interval is $7.9322 \pm 1.984(0.3803) = [7.1776, 8.6868]$.

A new occupation's outcome also varies around its process mean. Write $\mu_0 = \beta_0 + \beta_1 x_0$, $\hat{\mu}_0 = \hat{y}_0$, and $Y_0 = \mu_0 + \varepsilon_0$. The prediction error has two parts:

$$Y_0 - \hat{\mu}_0 = \varepsilon_0 - (\hat{\mu}_0 - \mu_0).$$

If this new error is independent of the fitted data and comes from the same common-variance process, the two variance contributions add:

$$\mathrm{Var}(Y_0 - \hat{\mu}_0 \mid X, x_0) = \sigma^2 + \mathrm{Var}(\hat{\mu}_0 \mid X, x_0).$$

The first term is new outcome variation; the second is uncertainty in the estimated mean. Substituting the fitted-mean variance and estimating σ by s_e gives

$$SE_{\text{new}} = s_e\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = 3.5041.$$

Thus the 95% prediction interval is approximately $7.9322 \pm 1.984(3.5041) = [0.9802, 14.8842]$. The extra 1 is the new outcome's own error variance, expressed as a multiple of σ^2; it does not disappear as the estimation sample grows.

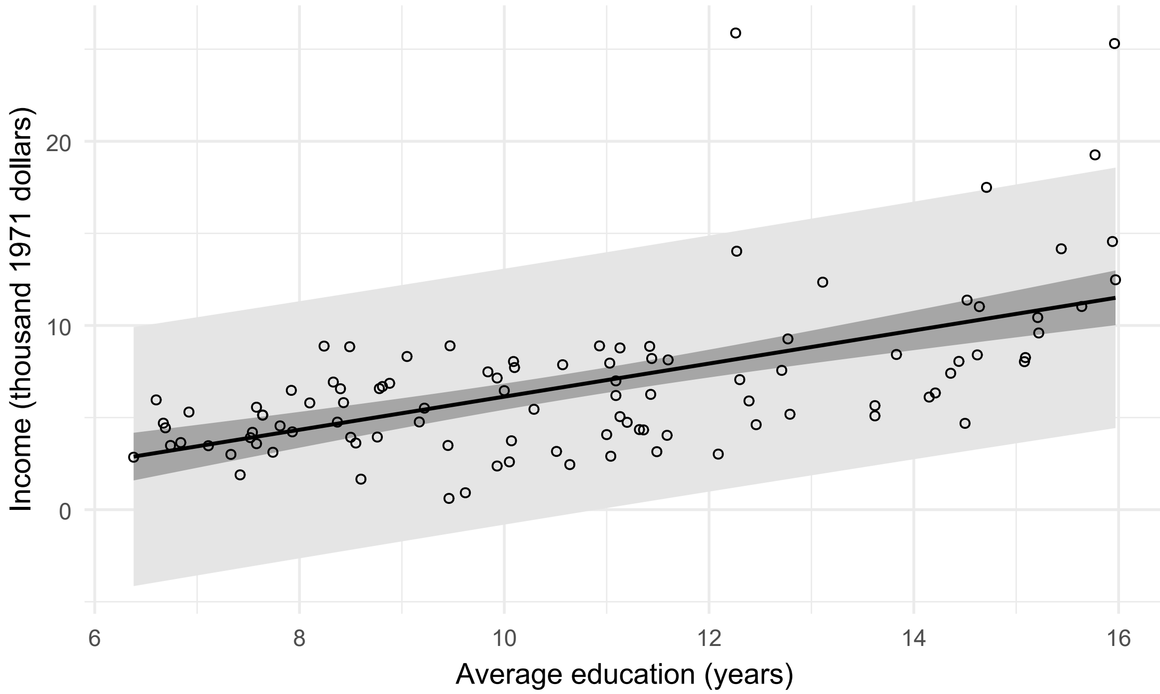

Figure 6: Mean-response intervals and new-case prediction intervals under the same working model.

Both intervals get wider away from $\bar{x}$, even within the observed range. Outside that range, numerical output is not a warrant for extrapolation. A lower prediction bound below zero also exposes the limitations of an unrestricted normal-error model for a nonnegative outcome; it is not a plausible negative-income forecast.

```
confint(model)
predict(model, data.frame(education = 12), interval = "confidence")
predict(model, data.frame(education = 12), interval = "prediction")
```

A separately owed new-case prediction answer is a separable question, not automatically another strand of the original descriptive question. Moreover, a model prediction interval is not a test of a learning algorithm on unseen records. Held-out prediction performance is a distinct estimand, developed with multiple regression. Appendix A retains the interval derivations and further supported-value comparisons.

7 Check Form, Spread, Influence, and Support

7.1 Checks should change an interpretation or a decision

The fitted line does not have to satisfy a stochastic model to exist. But claims that it describes a linear conditional mean, that its classical interval has exact coverage, or that it predicts new occupations reliably need additional support.

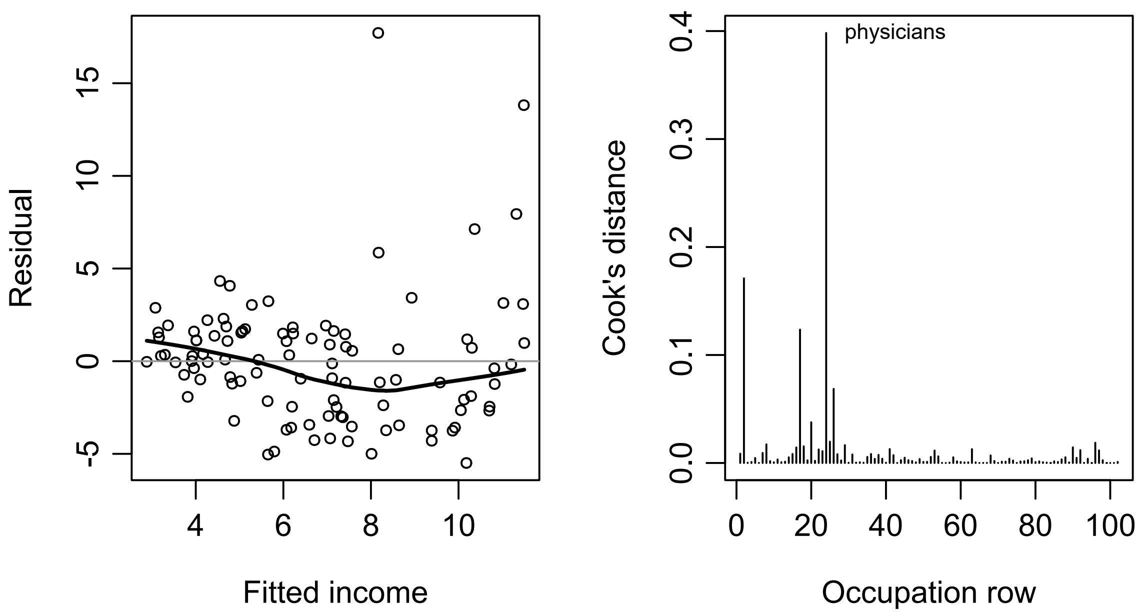

Figure 7: Residual pattern and influence invite different follow-up questions.

A curve in residuals suggests that one straight conditional mean leaves systematic variation. A changing residual spread challenges a common-variance uncertainty calculation. Leverage concerns unusual predictor positions; influence concerns how much the fitted model changes when a record's contribution changes. A large residual, high leverage, and high influence are related but different properties.

For a straight line with an intercept, leverage is

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

It grows as a predictor value moves away from $\bar{x}$. Cook's distance then combines leverage and residual magnitude:

$$D_i = \frac{e_i^2}{2s_e^2}\frac{h_i}{(1-h_i)^2}.$$

Cook's distance summarizes how much the fitted values change when case i is omitted, relative to residual variation and the two fitted coefficients. It is a prompt to inspect provenance and sensitivity, not an automatic deletion cutoff.

For these data, adding education squared lets the fitted relationship bend. The following same-record comparison makes the change in shape visible. It is an exploratory check, not a retrospectively prespecified new final model.

```
quad <- lm(
  income_k ~ education + I(education^2), data = prestige
)
most <- which.max(cooks.distance(model))
without <- lm(income_k ~ education, data = prestige[-most, ])
```

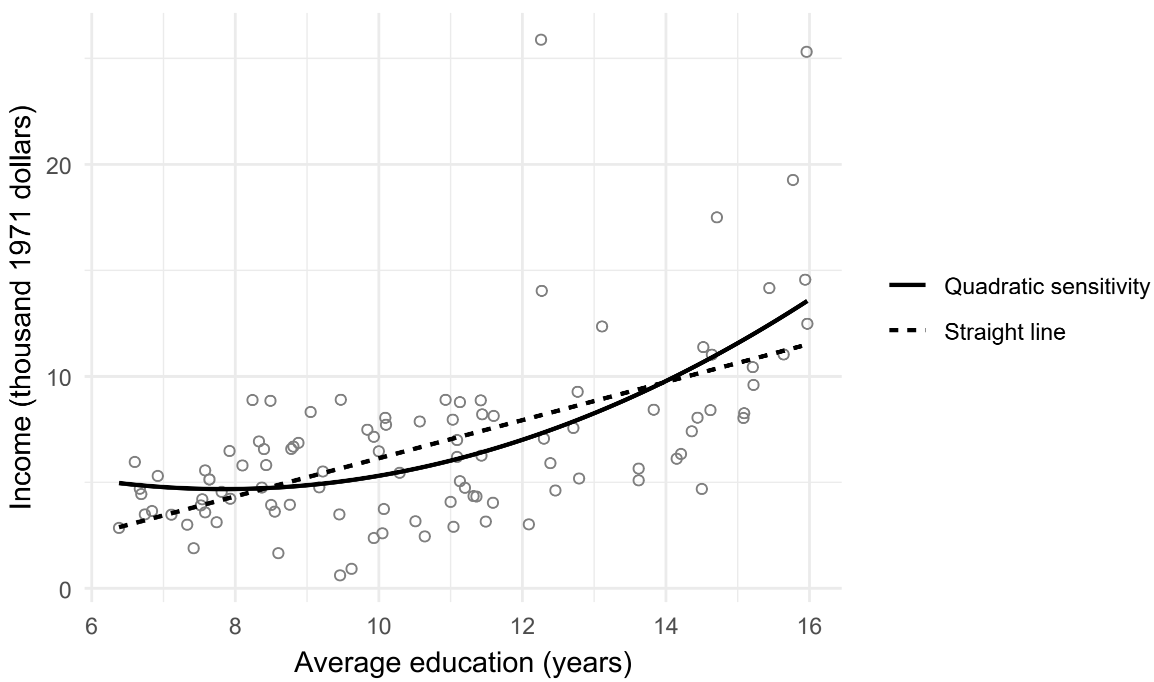

Figure 8: Straight and quadratic fits to the same 102 records. The curve is an exploratory sensitivity, not a selected final model.

The quadratic coefficient has $p = .0059$ under that model's classical reference calculation. This is a reason to examine the visible curvature, not a sufficient model-selection rule: unequal spread and choosing the comparison after inspecting the records also qualify that number.

Removing physicians, the record with the largest Cook's distance (0.398), changes the slope from 0.8988 to 0.7982. Nothing in that result proves physicians were recorded incorrectly. The deletion is a sensitivity calculation; it is not our replacement analytic sample or a licence to hide a valid unusual occupation.

An HC3 heteroskedasticity-consistent covariance calculation gives a slope SE of 0.1491 rather than 0.1270. Using the same t_{100} critical value as an approximate sensitivity procedure gives $[0.6031, 1.1945]$. The line does not change. This procedure relaxes a variance assumption under suitable independent-observation conditions; it does not repair curvature, dependence, confounding, or measurement error (MacKinnon & White, 1985; White, 1980). Its finite-sample interval is not the exact classical normal-model interval. The short R calculation makes the covariance choice explicit:

```r
V_hc3 <- sandwich::vcovHC(model, type = "HC3")
c(classical = coef(summary(model))["education", "Std. Error"],
  HC3 = sqrt(V_hc3["education", "education"]))
```

```
classical       HC3
   0.1270    0.1491
```

Appendix B retains the matrix sandwich construction and the optional best-linear-projection target under curvature.

Table 7
Diagnostic questions and their consequences.

Check	What needs to be examined	Consequence
Form	Residual curve and a curved-fit comparison	A straight line may remain a description but fail as a constant conditional-mean slope account
Spread	Residual variation over fitted values	Classical SEs may need revision; HC3 is a variance sensitivity, not a universal repair
Influence	Source record and result with/without a consequential case	Disclose sensitivity; correct errors only when verified
Independence	How occupations and measurements were generated or related	A plot cannot establish independent process realizations
Support	Actual education range and density of observations	Restrict comparisons; avoid extrapolation and individual-level claims

8 Read the Reported Argument, Then Evaluate Its Quality

8.1 Read the reported question and answer

To apply Chapter 1, we need a report, not merely an output table. The following **chapter-authored teaching report** is the accessible record. Its numbered paragraphs provide trace locations. They are not quotations from the original data source.

> ℹ Teaching report: an occupation-level description
>
> **R1 – Question and evidence.** We described the education–income relationship in the 102 *carData::Prestige* occupation records. Income was expressed in thousand 1971 Canadian dollars. All records had both modeled measurements. Each occupation was weighted equally; the observed education range was 6.38–15.97 years.
> **R2 – Offered reasons.** The upward scatterplot motivated a straight-line summary. OLS was chosen because it minimizes squared vertical discrepancies. The aim was a summary of these records, not a constant causal effect or an individual-level comparison. The source documentation identifies the measurements as historical occupational averages.

No probability-sampling justification was available in the record consulted.
R3 – Analysis and interpretation. The fitted slope was 0.8988 thousand dollars per education year, with $R^2 = .3336$ and residual SD 3.4834 thousand dollars. The residual pattern and a curved-fit sensitivity qualified the line as an incomplete description. The fit and sensitivities were exploratory. The descriptive result itself was not assigned sampling uncertainty.
R4 – Reported answer. Across these 102 recorded occupations, higher average education accompanied higher average income overall. The least-squares line summarized that pattern by about 899 additional 1971 dollars of fitted occupation-average income per education year over the observed range. It did not describe every occupation closely or establish an individual or causal return to education.

The report deliberately separates its descriptive conclusion from the hypothetical inferential route in Sections 4–6. A reader should not attach the model-based interval to R4 as though R4 named a population slope.

8.2 Locate the component content and its traces

Apply Chapter 1's functions to the report's actual content. A documentary trace identifies where that content can be inspected; it does not establish that the content is adequate.

Table 8
Reading one argument with linked statistical and substantive layers.

Function	Content read in this report	Trace and documentary status
STAT-E	Records, measurements, scaling, and complete sample	R1; stated in the paper
STAT-J	Reasons for OLS as a descriptive, noncausal summary	R2; stated in the paper
STAT-I	Slope, fit, sensitivity, and exploratory status	R3; stated in the paper
STAT-C	Positive but incomplete description over recorded support	R4; stated in the paper
SUB-E	Historical occupation averages, not individual measurements	R1 and data note; stated in the paper and located in related materials
SUB-J	Measurement meanings supporting an occupation-level reading	R2 and data note; stated in the paper and located in related materials

Function	Content read in this report	Trace and documentary status
SUB-I	Meaning of about 899 dollars per education year	R3–R4; stated in the paper
SUB-C	Bounded historical answer, without a causal or individual return	R4; stated in the paper

Here "the paper" means the supplied teaching report. Chapter 1's statuses remain **stated in the paper**, **located in related materials**, **implicit but traceable**, and **not located**; flag conflicts separately. Not located does not mean absent everywhere. One passage may serve several functions. The focal slope uses one statistical strand; fit summaries and checks support that strand rather than automatically becoming separately owed answers.

For a faulty R2 variant, consider "We used R, so our slope is an unbiased causal effect." Its offered reason remains located STAT-J content, not "not located." Reading records that reason and the reported answer; evaluation judges whether the reason supports the claim. Do not silently replace the report's answer with a better one.

8.3 Evaluate the argument already read

Table 9
Evaluating the argument already read.

Quality-evaluation question	Application to the reported descriptive argument
Question alignment	Does the answer remain about the 102 occupations and the declared straight-line summary?
Evidence integrity	Are the source, aggregate unit, scaling, exclusions, and unusual records traceable?
Justification adequacy	Are the offered reasons adequate for a descriptive line, rather than stretched into random sampling or causal identification?
Inferential adequacy	Do the calculation and substantive reasoning support the reported pattern while retaining curvature and scatter?
Transition fidelity	Does the interpretation preserve occupation-level measurements, units, and exploratory status?

Quality-evaluation question	Application to the reported descriptive argument
Claim calibration	Does the answer retain historical scope, supported range, and the limits of a linear summary?

These are six questions, not six components or a total score. A local judgment can be adequate for this purpose, needs revision, insufficient information to judge, or not applicable. The judgment belongs to evaluation, not to the documentary-status column of the reading table.

For the deliberately limited R4 answer, the overall conclusion is **supported as stated**: it reports a checkable finite-record summary and acknowledges what the line does not represent. Missing details about national sampling do not invalidate arithmetic on the named records, but would matter for a national generalization.

The additional model-based extension is less secure. Its exact classical uncertainty justification is not established merely by calculating an interval, and the diagnostics challenge its linear-mean and common-variance conditions. Retaining the descriptive answer does not rescue that distinct process claim. A stochastic best-linear-projection slope under curvature would be another explicitly defined estimand with suitable random-X conditions and uncertainty, not an unannounced relabelling of β_1.

The opening claim about an individual's return to education is **not supported** by this record. Replacing it with R4 changes the unit and mode of the question; that is a revised question–answer pair, not cosmetic narrowing of the same claim. A causal comparison needs different support, not a different software command.

In Project STAR (Finn & Achilles, 1990; Mosteller, 1995), a binary-regression slope equals a difference in observed class-condition means. That arithmetic alone supplies no causal warrant. The within-school assignment design matters, together with the link from recorded class type to original assignment, missing outcomes, school weights, and dependence among pupils. This teaching reanalysis does not independently verify the assignment link. Schools are assignment blocks and covariance clusters, not the units assigned to class conditions. The full Chapter 8, Section 8: Project STAR works through these distinctions and separates the observed-reading sample from the smaller covariate-complete sample. It is a different question, not a prerequisite for the bounded occupation answer.

8.4 Auxiliary construction callback

For this regression, **Understand and plan** would declare the occupation unit, historical scope, finite-record slope, variables, and proposed checks before fitting. **Produce and complete** would preserve the source record, estimate and check the line, then report the bounded occupation-level answer. This is a construction callback, not a reading sequence. Revising uncertainty for the same question may repair Phase 2; changing the unit, outcome, mode, or estimand starts a new or revised cycle. A retrospective reanalysis does not become prospective merely because the workflow is described afterward.

9 Summary and Reading Bridge

A simple regression starts with a comparison and a unit, not a coefficient table. Its fitted values and residuals are calculations on records. OLS connects the slope to covariance and predictor variation; its intercept places the line through the two means. R^2, residual SD, slope uncertainty, and new-case variation answer different questions.

Across the 102 recorded occupations, the fitted line rises by 0.8988 thousand 1971 Canadian dollars—about 899 dollars—for each additional year of occupation-average education over the observed 6.38–15.97-year range. This is a bounded finite-record summary, not an individual, causal, or population return.

Known-truth repetition shows why fitted slopes vary, what their standard error estimates, and why 95% coverage belongs to a procedure rather than to one fixed parameter after the data are observed. Noise, predictor spread, sample size, and dependence affect precision in different ways.

Reading an argument locates the offered support and reported answer. Quality evaluation then asks whether that support is adequate for the declared claim. The auxiliary workflow explains how such an analysis and record can be constructed. None of these activities permits a finite-record slope to acquire a population interpretation simply because software printed a standard error.

For another accessible account of lines, residuals, and fit, see OpenIntro's single-predictor regression tutorial. For a compact executable route from a fitted model to intervals and checks, see the ISLR authors' Chapter 3 R lab. These provide complementary practice; the present chapter additionally makes the finite-record and model-based questions explicit. Consult the carData reference manual for data documentation, and Fox & Weisberg (2019) and Gelman et al. (2020) for broader regression treatments.

Chapter 8 next asks conditional-comparison and coordinated association/prediction questions. It begins adjustment by refitting the education-only model on the same 98 type-complete occupation records: moving from 102 to 98 is a sample change, not an adjustment effect. A short indicator-variable bridge connects regression to group comparisons; fuller ANOVA algebra stays optional. Interaction, where one comparison depends on another predictor, receives its own later treatment.

Exercises

The prompts below use **Foundational** and **Integrative** levels. Where rounded summaries are supplied, answers within 0.02 in the stated units are sufficient. No optional appendix is needed for Exercises 1–6.

Exercise 1: Read the unit and the line (Foundational)

A teaching report describes 80 occupations with $\widehat{\text{income}} = -1.5 + 0.7\,\text{education}$, where income is in thousand dollars and average education ranges from 7 to 16 years. Interpret the slope and the intercept. Calculate the fitted income at 12 years and the residual for an occupation at 12 years with income 9.0. Explain why the slope does not give an individual's gain from another year of schooling.

Exercise 2: Connect covariance and the fitted line (Foundational)

A set of records has $\bar{x} = 4, \bar{y} = 10, S_{xx} = 20, S_{xy} = 30$. Calculate the OLS slope and intercept, and the fitted value at $x = 5$. Show that the line passes through the mean point. If $s_x = 2$ and $s_y = 5$, calculate the correlation implied by these summaries. What would prevent fitting a slope if every x were equal?

Exercise 3: Separate fit and uncertainty (Foundational)

An intercept-and-slope regression has $n = 22$, $RSS = 80$, $TSS = 144$, $S_{xx} = 100$, and fitted slope 0.8. Calculate residual SD, R^2, and the classical slope SE. Use $t_{.975,20} = 2.086$ to obtain a 95% slope interval. State the process assumptions needed for that interval and explain why it does not by itself establish that the line fits closely or identifies a cause.

Exercise 4: Mean or new case? (Foundational)

Under the stated independent normal-error constant-variance model, a fitted value at $x_0 = \bar{x}$ is 12, $n = 25$, and $s_e = 3$. Use $t_{.975,23} = 2.069$. Calculate the SE and 95% interval for the conditional mean, then the corresponding prediction SE and interval for one independent new case at the same x_0. Explain the extra 1 in the prediction expression. Would either interval validate the model in a different population?

Exercise 5: Locate first, evaluate second (Integrative)

A report states: "We used a regression program, so the coefficient is a causal effect. Our 60 records are school averages. The fitted slope is 4 test-score points per additional average study hour. The observed study-hour range is 2–6." No assignment or sampling documentation is provided.

Identify an offered justification and its documentary status. State the reported claim before evaluating it. Apply Justification adequacy and Transition fidelity. Explain whether an individual causal conclusion is supported, and identify a narrower descriptive question that the fitted line could address without pretending it was the original causal question.

Exercise 6: Respond to sensitivity (Integrative)

In a descriptive analysis of 102 occupations, the slope is 0.90. A valid, high-income occupation has the largest Cook's distance; removing it gives 0.79. A curved-fit check changes the shape, and an HC3 calculation changes the slope SE from 0.13 to 0.15 without changing the line.

What does each check tell you? What does it not tell you? Explain why deleting the occupation automatically is inappropriate. Distinguish a variance repair for the same estimand from replacing a constant process slope with a different estimand. Write a short sensitivity disclosure for the descriptive analysis.

Appendix A: Extended regression mathematics and interval derivations

The main text uses the identities needed to read a fitted line. This appendix derives them and distinguishes identities about the observed records from claims about hypothetical repetitions. All occupation calculations retain the same 102 rows, education in years, and income in thousand 1971 Canadian dollars. Recreating these objects makes the appendix code usable independently of the main-text examples.

```
ch7app_data <- carData::Prestige
ch7app_data$occupation <- rownames(ch7app_data)
ch7app_data$income_k <- ch7app_data$income / 1000
ch7app_model <- lm(income_k ~ education, data = ch7app_data)
ch7app_x <- ch7app_data$education
ch7app_y <- ch7app_data$income_k
ch7app_n <- nrow(ch7app_data)
ch7app_Sxx <- sum((ch7app_x - mean(ch7app_x))^2)
ch7app_Sxy <- sum(
```

```
  (ch7app_x - mean(ch7app_x)) * (ch7app_y - mean(ch7app_y))
)
```

Derivatives, normal equations, and the minimizing line

For candidate intercept b_0 and slope b_1, define

$$Q(b_0, b_1) = \sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2.$$

Squared residuals cannot cancel and give large vertical discrepancies greater weight. Differentiating with respect to each candidate coefficient gives

$$\frac{\partial Q}{\partial b_0} = -2\sum_i (y_i - b_0 - b_1 x_i), \qquad \frac{\partial Q}{\partial b_1} = -2\sum_i x_i(y_i - b_0 - b_1 x_i).$$

At the minimum both derivatives are zero. The resulting **normal equations** are

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i = \sum_i y_i, \qquad \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2 = \sum_i x_i y_i.$$

The first equation gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Substitution into the second yields

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = \frac{S_{xy}}{S_{xx}}.$$

Here $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$ is required: a slope cannot be identified when every predictor value is the same. Because sample covariance and variance use the same denominator $n - 1$,

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

In the occupation data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} = 0.8988, \qquad \hat{\beta}_0 = 6.7979 - 0.8988(10.7380) \approx -2.8534.$$

The ratio has units of thousand dollars per education year. Unlike a correlation, the slope changes when a variable's units change. Standardizing both variables makes their sample standard deviations one, so the simple intercept-model slope becomes r.

To verify that the stationary point is the unique minimum, write any other line as $\hat{\beta}_0 + d_0 + (\hat{\beta}_1 + d_1)x_i$. The normal equations make the cross term with the OLS residuals vanish, leaving

$$Q(\hat{\beta}_0 + d_0, \hat{\beta}_1 + d_1) = RSS + n(d_0 + d_1\bar{x})^2 + d_1^2 S_{xx}.$$

With $S_{xx} > 0$, every different line has larger RSS. If a candidate line is constrained to pass through $(\bar{x}, \bar{y})$, its excess RSS is simply $S_{xx}(b_1 - \hat{\beta}_1)^2$. The numbered method route makes this comparison visible with five candidate lines.

Residual constraints and projection geometry

At the OLS solution the derivative equations become $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Thus the residuals have zero sample covariance with the predictor and mean zero **by construction**. They are not evidence that the unknown population errors are independent, normally distributed, or conditionally mean zero.

For the geometry, let $\mathbf{X}$ be the $n \times 2$ matrix with first column all ones and second column containing the predictor. Bold $\mathbf{X}$ is a matrix, distinct from a single predictor value x_i. The normal equations can be written

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}, \qquad \hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

The fitted vector is the projection of $\mathbf{y}$ onto the two-dimensional space spanned by the intercept and predictor columns. The residual vector is perpendicular to that space. Writing $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, we have $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

Because the model includes an intercept, both $\hat{\mathbf{y}}$ and the constant vector $\bar{y}\mathbf{1}$ lie in the fitted space. Consequently,

$$\begin{aligned}\sum_i (y_i - \bar{y})^2 &= \sum_i \{(\hat{y}_i - \bar{y}) + e_i\}^2 \\ &= \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i e_i^2,\end{aligned}$$

since $\sum_i (\hat{y}_i - \bar{y})e_i = 0$. This proves $TSS = RegSS + RSS$, an algebraic decomposition of observed variation, not a division into causal contributions. For these records, $1820.8134 = 607.4214 + 1213.3920$, in squared thousand-dollar units.

Also $RegSS = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$. Dividing by $TSS = S_{yy}$ proves $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$ for simple regression **with an intercept**. The identities need not hold for a no-intercept fit. When fitted values are nonconstant, $\operatorname{cor}(y, \hat{y})^2 = R^2$ as well.

There are two fitted directions and $n-2$ residual directions. Under a correct linear mean with independent constant-variance errors, $E(RSS \mid \mathbf{X}) = (n-2)\sigma^2$. Thus $s_e^2 = RSS/(n-2)$ estimates the error variance, while s_e estimates its standard deviation in the outcome's units. The square root is not itself an exactly unbiased estimator of σ. Without that stochastic model, s_e remains a computable residual-scale summary, not proof of a common population error variance.

From an error model to slope and intercept uncertainty

The 102-record least-squares line is already determined by those records. The next derivation concerns a **separate model-based question**: what constant conditional-mean coefficient generated hypothetical independent occupation outcomes at the observed education values? Assume

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad E(\varepsilon_i \mid \mathbf{X}) = 0, \quad \mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}.$$

These are offered working conditions, not consequences of the package's availability or of fitting `lm()`. The slope error is

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x})\varepsilon_i}{S_{xx}}.$$

Independence and common variance then give

$$\mathrm{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma^2}{S_{xx}}, \qquad SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{S_{xx}}}.$$

Because $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{Cov}(\bar{Y}, \hat{\beta}_1 \mid \mathbf{X}) = 0$,

$$\mathrm{Var}(\hat{\beta}_0 \mid \mathbf{X}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \qquad \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

This covariance explains why simply adding the variances of an uncentered intercept and slope prediction would be wrong. If the predictor is centered at its observed mean, the intercept and slope estimates are uncorrelated under the classical model, and the intercept SE is $s_e/\sqrt{n}$.

Under the stated linear-mean, uncorrelated, equal-variance conditions, OLS is the best linear unbiased estimator: it has the smallest variance among estimators linear in the outcomes and unbiased for the coefficients. This Gauss–Markov statement does not require normal errors and does not establish causal identification, valid measurement, or population coverage (Fox, 2016; Kutner et al., 2005).

Add a jointly normal error vector conditional on the predictor values. Then the standardized coefficient error is normal, RSS/σ^2 has a χ^2_{n-2} distribution, and these quantities are independent. Their ratio gives

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-2}, \qquad j = 0, 1.$$

That is the basis of exact classical conditional coverage for $\hat{\beta}_j \pm t_{.975,n-2} SE(\hat{\beta}_j)$, and of the corresponding null test. For the occupation slope the calculation is $0.8988 \pm 1.9840(0.1270) = [0.6468, 1.1508]$ thousand dollars per year. It illustrates the model-based procedure; the diagnostics in Section 7 do not support treating its nominal exact coverage as established here.

Fixed-X derivations condition on the predictor values; those values need not literally have been experimentally fixed. A random-X conditional-mean model can use the same conditional argument when its error conditions hold. Neither route demonstrates that these occupations were probability sampled. Normality of the marginal distribution of education or income is not required. The assumptions concern conditional errors, not the two marginal histograms.

Why a new-case interval has an additional variance term

For a supported value x_0, define the model mean $\mu_0 = \beta_0 + \beta_1 x_0$ and its estimate $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Substituting the coefficient variances and covariance above gives

$$\begin{aligned} \mathrm{Var}(\hat{\mu}_0 \mid \mathbf{X}) &= \mathrm{Var}(\hat{\beta}_0) + x_0^2 \,\mathrm{Var}(\hat{\beta}_1) + 2x_0 \,\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left\{ \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}. \end{aligned}$$

The common conditioning on $\mathbf{X}$ is suppressed in the first line. A confidence interval for this mean uses the square root of this expression with s_e replacing σ.

A new occupation outcome is instead $Y_0 = \mu_0 + \varepsilon_0$. If its error is independent of the fitted data and comes from the same normal, equal-variance process,

$$\mathrm{Var}(Y_0 - \hat{\mu}_0 \mid \mathbf{X}, x_0) = \sigma^2 \left\{ 1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}.$$

The extra 1 is new outcome variation; it does not disappear even if the model mean were known. Both standard errors use the same $t_{.975,n-2}$ critical value for their classical 95% intervals under the full model.

```r
ch7app_new <- data.frame(education = c(8, 10, 12, 15))
ch7app_mean_intervals <- predict(
  ch7app_model, newdata = ch7app_new, interval = "confidence"
)
ch7app_case_intervals <- predict(
  ch7app_model, newdata = ch7app_new, interval = "prediction"
)
ch7app_mean_intervals
ch7app_case_intervals
```

The table puts both sets of limits beside the common fitted values.

Education	Fitted	Mean_lower	Mean_upper	Case_lower	Case_upper
8	4.34	3.37	5.31	-2.64	11.32
10	6.13	5.43	6.84	-0.81	13.08
12	7.93	7.18	8.69	0.98	14.88
15	10.63	9.36	11.90	3.60	17.66

At 10 years, the point prediction is 6.13 thousand dollars, the mean interval is [5.43, 6.84], and the new-occupation interval is [-0.81, 13.08]. The negative endpoint exposes the lower-tail limitation of this unbounded normal-linear model. Cutting it off at zero does not produce a validated 95% interval. An outcome transformation or a model respecting the boundary would change the analysis and would need its own interpretation and checks.

Both intervals widen away from $\bar{x}$; their narrowest point is the mean education value. With more comparable independent observations and stable predictor spread, mean uncertainty tends to shrink. The new-case interval retains a nonzero error-variation floor. None of these formulas includes uncertainty from an inappropriate mean function, biased measurement, or a shift to another labour market. They are pointwise model intervals, not a simultaneous band or empirical out-of-sample validation. Their target is a new occupation-level record from the declared process, never an individual worker's income.

Appendix B: Extended covariance and projection targets

This appendix retains optional matrix and target-level extensions. The known-truth and repeated-sample demonstrations now appear in the numbered method route, where they explain the ordinary slope standard error and coverage interpretation. Nothing programmed in those simulations validates the empirical occupation model.

Unequal variance and the HC3 sandwich calculation

For independent observations, a heteroskedasticity-consistent covariance estimate allows the error variance to differ across predictor values. It changes estimated coefficient uncertainty, not the OLS coefficients. Let $\mathbf{x}_i = (1, x_i)^\mathsf{T}$ denote a design column for case i, and let $h_i = \mathbf{x}_i^\mathsf{T}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{x}_i$ be its leverage. In simple regression with an intercept,

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

HC3 uses the leverage-adjusted squared residual $w_i = e_i^2/(1-h_i)^2$ inside

$$\widehat{\mathrm{Var}}_{HC3}(\hat{\beta}) = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\left(\sum_i w_i \mathbf{x}_i \mathbf{x}_i^\mathsf{T}\right)(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}.$$

The two outer factors are often called the sandwich's bread and the middle factor its meat. The original heteroskedasticity-consistent construction uses case-specific squared residuals; HC3 inflates contributions from high-leverage cases to address finite-sample behavior (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980). It does not reveal each case's unknown true variance.

```r
ch7app_X <- model.matrix(ch7app_model)
ch7app_e <- resid(ch7app_model)
ch7app_h <- hatvalues(ch7app_model)
ch7app_w <- (ch7app_e / (1 - ch7app_h))^2
ch7app_bread <- solve(crossprod(ch7app_X))
ch7app_meat <- t(ch7app_X) %*% (ch7app_X * ch7app_w)
ch7app_vcov_hc3 <- ch7app_bread %*% ch7app_meat %*% ch7app_bread
ch7app_se_hc3 <- sqrt(diag(ch7app_vcov_hc3))
ch7app_critical <- qt(0.975, df.residual(ch7app_model))
ch7app_hc3_ci <- cbind(
  lower = coef(ch7app_model) - ch7app_critical * ch7app_se_hc3,
  upper = coef(ch7app_model) + ch7app_critical * ch7app_se_hc3
)
c(
  classical_slope_SE = coef(summary(ch7app_model))["education", "Std. Error"],
  HC3_slope_SE = ch7app_se_hc3["education"]
)
```

```
classical_slope_SE HC3_slope_SE.education
            0.1270                 0.1491
```

```r
ch7app_hc3_ci["education", ]
```

```
 lower  upper
0.6031 1.1945
```

The slope SE changes from 0.127 to 0.149 thousand dollars per year. Using the same t_{100} critical value yields [0.603, 1.195], an **approximate sensitivity interval**, not an exact normal-theory t interval under heteroskedasticity. A maintained implementation such as `sandwich::vcovHC(ch7app_model, type = "HC3")` is preferable in applied work once the calculation is understood (Zeileis, 2004).

The comparison challenges the common-variance uncertainty calculation. HC3 does not correct a nonlinear mean, dependent occupations, confounding, or errors in measurement. Under the classical measurement-error setup, independent noise added to a true predictor attenuates the population simple-regression slope, while independent mean-zero outcome noise increases error variation without changing that slope (Cohen et al., 2003; Fox, 2016). Residual plots cannot recover the true predictor or establish measurement validity.

Diagnostic limits and a projection target

The numbered route now contains the ordinary curvature, leverage, influence, and HC3 sensitivity work. Its same-record comparison preserves the original outcome scale and analytic sample. The appendix keeps the more technical target that becomes relevant if a constant conditional-mean slope is abandoned.

Neither a residual plot nor a quadratic comparison can certify independent occupations. Dependence requires evidence about how occupational categories and their measurements were produced. A quantile plot can flag tail behavior relevant to normal-theory inference, but cannot establish the sampling mechanism. Extrapolation is likewise a support problem: the observed education range is 6.38–15.97 years. A line can return a value at 20 years without making that value empirically supported.

Curvature leaves the finite-record OLS slope well defined but challenges its interpretation as a constant conditional-mean contrast. A **best linear projection** is a distinct stochastic target,

$$(\beta_0^*, \beta_1^*) = \arg\min_{a,b} E\{(Y - a - bX)^2\}, \qquad \beta_1^* = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)},$$

when the required moments exist and $\mathrm{Var}(X) > 0$. Its residual is orthogonal to 1 and X in the population, but need not have zero mean at every $X = x$. With independent identically distributed pairs and suitable moment conditions, robust covariance methods can support asymptotic inference for that projection target. Such a target depends on the joint distribution of occupations and education and is not a constant conditional-mean derivative. It must be declared as a new or revised model-based question; it cannot retroactively receive an exact classical interval from the rejected linear-mean model. The descriptive conclusion about the observed least-squares line does not require making that extension.

References

Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences* (3rd ed.). Lawrence Erlbaum Associates.

Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, *27*(3), 557–577.

Fox, J. (2016). *Applied regression analysis and generalized linear models* (3rd ed.). SAGE Publications.

Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. https://www.john-fox.ca/Companion/index.html

Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. https://doi.org/10.32614/CRAN.package.carData

Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and other stories*. Cambridge University Press. https://doi.org/10.1017/9781139161879

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.

Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, *54*(3), 217–224. https://doi.org/10.1080/00031305.2000.10474549

MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, *29*(3), 305–325. https://doi.org/10.1016/0304-4076(85)90158-7

Mosteller, F. (1995). The Tennessee study of class size in the early school grades. *The Future of Children*, *5*(2), 113–127.

Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, *15*(3), 351–357. https://doi.org/10.2307/2087176

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, *48*(4), 817–838. https://doi.org/10.2307/1912934

Zeileis, A. (2004). Econometric computing with HC and HAC covariance matrix estimators. *Journal of Statistical Software*, *11*(10), 1–17. https://doi.org/10.18637/jss.v011.i10